Probabilistic Combinatorics

Sheet 11

July 1, 2024

Problem 1. Consider Erdős-Rényi model $G(n,p)$. Using the Janson-inequality show that the probability that a graph is triangle free is bounded from above by $e^{-\mu+\delta/2}$ for $\mu = \binom{n}{3}p^3$ and $\Delta = 3\binom{n}{3}(n-3)p^5$.

Proof.
It is known that the Erdős-Rényi model $G(n,p)$ is used for randomly generating undirected graph of n vertices where the probability to choose an edge is p. Let Ω be the set of all possible edges between n vertices. In this problem, we consider the discrete probability space $(2^\Omega, \mathbb{P})$ where the probability measure $\mathbb{P}(E) = \prod_{e\in E} p \prod_{e\notin E}(1-p)$ for any set of edges $E \in 2^\Omega$. Consider the events $A_1, \ldots, A_{\binom{n}{3}} \subseteq \Omega$, which are 3-elements sets where each edge-set A_i forms a triangle. For $1 \le i \le \binom{n}{3}$, let $B_i = A_i$ and X_i be the indicator random variable for B_i, and $X = X_1 + X_2 + \ldots + X_{\binom{n}{3}}$. It is obvious that $\mathbb{P}(B_i) = p^3$ as three edges in the edge-set B_i must appear to form a triangle, and the edges are chosen independently with probability p in the random graph model $G(n,p)$. By linearity of expectation, we have

$$\mu = \mathbb{E}(X) = \mathbb{E}(\sum_{i=1}^{\binom{n}{3}} X_i) = \sum_{i=1}^{\binom{n}{3}} \mathbb{E}(X_i) = \sum_{i=1}^{\binom{n}{3}} \mathbb{P}(B_i) = \binom{n}{3}p^3$$

Next, we want to calculate

$$\Delta = \sum_{\substack{A_i\cap A_j\neq\emptyset \\ 1\le i,j\le\binom{n}{3}, i\neq j}} \mathbb{P}(B_i \cap B_j) \tag{1.1}$$

For any $i \neq j$ where $1 \le i, j \le \binom{n}{3}$, it is obvious that $|B_i \cap B_j| = 1$, as it is impossible to find B_i and B_j such that $|B_i \cap B_j| = 2$, and $B_i = B_j$ when $|B_i \cap B_j| = 3$. Such constraint implies that B_i and B_j have one edge in common, and thus $\mathbb{P}(B_i \cap B_j) = p^5$, as B_i and B_j are triangles having one common edge. That is, it remains to find the number of i, j, where $1 \le i, j \le \binom{n}{3}$ such that $i \neq j$, and $A_i \neq A_j$, in order to find Δ by the equation (1.1). For any i where $1 \le i \le \binom{n}{3}$, the procedure for constructing an A_j satisfying the constraint described above can be described as follows. First of all, we can select one of three edges in A_i as the common edge between A_i and A_j. For each selection, we want to choose a vertex that is not in A_i and draw an edge between this vertex and the endpoints of the common edge, and this indeed form a new triangle, which is A_j. That is, for any i where $1 \le i \le \binom{n}{3}$, there are $3(n-3)$ ways to find A_j satisfying the constraint described above. By the equation (1.1), we have

$$\Delta = 3\binom{n}{3}(n-3)p^5$$

With $B_1, \ldots, B_{\binom{n}{3}}, \mathbb{P}, X, \mu, \Delta$ described above, by the Janson Inequality, we have

$$\mathbb{P}\Big(\bigcap_{i=1}^{\binom{n}{3}}(2^\Omega \setminus B_i)\Big) = \mathbb{P}(X = 0) \le e^{-\mu+\Delta/2}$$

Since $\bigcap_{i=1}^{\binom{n}{3}}(2^\Omega \setminus B_i)$ is the event where the graph does not have any cycle, then the proof is finished. □

Problem 2. Consider the Erdős-Rényi model $G(n,p)$ of random graphs.

Using the results about correlation inequalities show that

$$\mathbb{P}(G(n,p) \text{ is triangle free}) \geq (1-p^3)^{\binom{n}{3}}$$

Proof.
It is known that the Erdős-Rényi model $G(n,p)$ is used for randomly generating undirected graph of n vertices where the probability to choose an edge is p. Let $r = \binom{n}{2}$, suppose that all possible edges between n vertices are numbered distinctly by using numbers in $[r]$. In this problem, we consider the discrete probability space $(2^{[r]}, \mathbb{P})$ where the probability measure $\mathbb{P}(E) = p^E(1-p)^{r-E}$ for any edge $E \in 2^{[r]}$. Consider the events $A_1, \ldots, A_{\binom{n}{3}} \subseteq [r]$, which are 3-elements sets where each edge-set A_i does not form a triangle. It it obvious that $\mathbb{P}(A_i) = 1-p^3$. For i where $1 \leq i \leq \binom{n}{3}$, let $\mathcal{A}_i = A_i$. It is obvious that $\mathbb{P}\Big(\bigcap_{i=1}^{\binom{n}{3}} \mathcal{A}_i\Big)$ is the probability that $G(n,p)$ is triangle-free. Based on the Corollary 12.8 in the lecture, we have

$$\mathbb{P}\Big(\bigcap_{i=1}^{\binom{n}{3}} \mathcal{A}_i\Big) \geq \prod_{i=1}^{\binom{n}{3}} \mathbb{P}(\mathcal{A}_i) = (1-p^3)^{\binom{n}{3}}$$

and the proof is finished. □

Problem 3. Let $B_1, \ldots, B_r, \mathbb{P}, X, \mu$, and Δ be like in the setup of the Janson-inequality. For every $T \subset [r]$, let

$$\begin{aligned} X_T &= \sum_{i \in T} X_i \\ \mu_T &= \mathbb{E}(X_T) \\ \Delta_T &= \sum_{\substack{i,j \in T, i \neq j \\ A_i \cap A_j \neq \emptyset}} \mathbb{P}(B_i \cap B_j) \end{aligned}$$

(i) Show that for every $T \subset [r]$ it holds, that

$$\mathbb{P}(X=0) \leq e^{-\mu_T + \frac{\Delta_T}{2}}$$

(ii) Now choose T randomly by independently including every $i \in [r]$ with probability q. Prove that:

$$\mathbb{E}(-\mu_T + \frac{\Delta_T}{2}) = -q\mu + \frac{q^2\Delta}{2}$$

Proof.
(i): In order to prove this inequality, we simply apply Janson Inequality to B_i for $i \in T$, μ_T, and Δ_T.

(ii): By linearity of expectation, we have

$$\mathbb{E}(-\mu_T + \frac{\Delta_T}{2}) = \mathbb{E}(-\mu_T) + \mathbb{E}(\frac{\Delta_T}{2}) \tag{3.1}$$

For $i \in [r]$, let $1_{i \in T}$ be the indicator function for the existence of i in T. It is obvious that $\mathbb{E}(1_{i \in T})$ is equal to the probability of i is in T, which is q. That is, we can rewrite X_T as follows

$$X_T = \sum_{i \in [r]} 1_{i \in T} X_i$$

By linearity of expectation and the fact that $1_{i\in T}$ and X_i are independent for $i \in [r]$, we have

$$\begin{aligned}\mathbb{E}(-\mu_T) &= -\sum_{i\in[r]} \mathbb{E}(1_{i\in T})\mathbb{E}(X_i) \\ &= -\sum_{i\in[r]} q\mathbb{E}(X_i) \\ &= -q\sum_{i\in[r]} \mathbb{E}(X_i) \\ &= -q\mu \end{aligned} \tag{3.2}$$

Similarly, we can rewrite Δ_T as follows

$$\begin{aligned}\Delta_T &= \sum_{\substack{i,j\in T, i\neq j \\ A_i\cap A_j\neq\emptyset}} \mathbb{P}(B_i \cap B_j) \\ &= \sum_{\substack{i,j\in [r], i\neq j \\ A_i\cap A_j\neq\emptyset}} 1_{i\in T}1_{j\in T}\mathbb{P}(B_i \cap B_j)\end{aligned}$$

By linearity of expectation and the fact that $1_{i\in T}$, $1_{j\in T}$, and $\mathbb{P}(B_i \cap B_j)$ are independent for $i \in [r]$, we have

$$\begin{aligned}\mathbb{E}(\frac{\Delta_T}{2}) &= \frac{1}{2}\sum_{\substack{i,j\in [r], i\neq j \\ A_i\cap A_j\neq\emptyset}} \mathbb{E}(1_{i\in T})\mathbb{E}(1_{j\in T})\mathbb{E}\Big(\mathbb{P}(B_i \cap B_j)\Big) \\ &= \frac{1}{2}\sum_{\substack{i,j\in [r], i\neq j \\ A_i\cap A_j\neq\emptyset}} q^2\mathbb{P}(B_i \cap B_j) \\ &= \frac{q^2}{2}\sum_{\substack{i,j\in [r], i\neq j \\ A_i\cap A_j\neq\emptyset}} \mathbb{P}(B_i \cap B_j) \\ &= \frac{q^2\Delta}{2}\end{aligned} \tag{3.4}$$

By the three equalities: $(3.1), (3.2), (3.3)$, the proof is finished. □

Problem 4. (Second Janson Inequality) Consider the setup of (3), but now also assume that $\Delta \geq \mu$. Use (3) to prove that:

$$\mathbb{P}(X = 0) \leq e^{\frac{-\mu^2}{2\Delta}}$$

Proof.

Based on the previous problem, we have $\mathbb{P}(X = 0) \leq e^{-\mu_T+\frac{\Delta_T}{2}}$ and $\mathbb{E}(-\mu_T + \frac{\Delta_T}{2}) = -q\mu + \frac{q^2\Delta}{2}$. By linearity of expectation, there must exist some choice of $T \in [r]$ such that

$$-\mu_T + \frac{\Delta_T}{2} \leq -q\mu + \frac{q^2\Delta}{2}$$

and thus

$$\mathbb{P}(X = 0) \leq e^{-q\mu+\frac{q^2\Delta}{2}} \tag{4.1}$$

Let $f(q) = \mathbb{E}(-\mu_T + \frac{\Delta_T}{2}) = -q\mu + \frac{q^2\Delta}{2}$. We have $f^{'}(q) = -\mu + q\Delta$. For $q = \frac{\mu}{\Delta}$, then $f^{'}(q) = 0$ and $f(q)$ is minimized. Since $\Delta \geq \mu$, then $q = \frac{\mu}{\Delta} \leq 1$, and thus $q = \frac{\mu}{\Delta}$ is valid as q is a probability measure. For such q, then the inequality (4.1) is equivalent to

$$\mathbb{P}(X = 0) \leq e^{\frac{-\mu^2}{2\Delta}}$$

and the proof is finished. □